Muscle Coactivation Levels Reflect a Tradeoff Between Metabolic Cost and Performance When Responding to Physical Disturbances During Upper Limb Posture Control

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Introduction

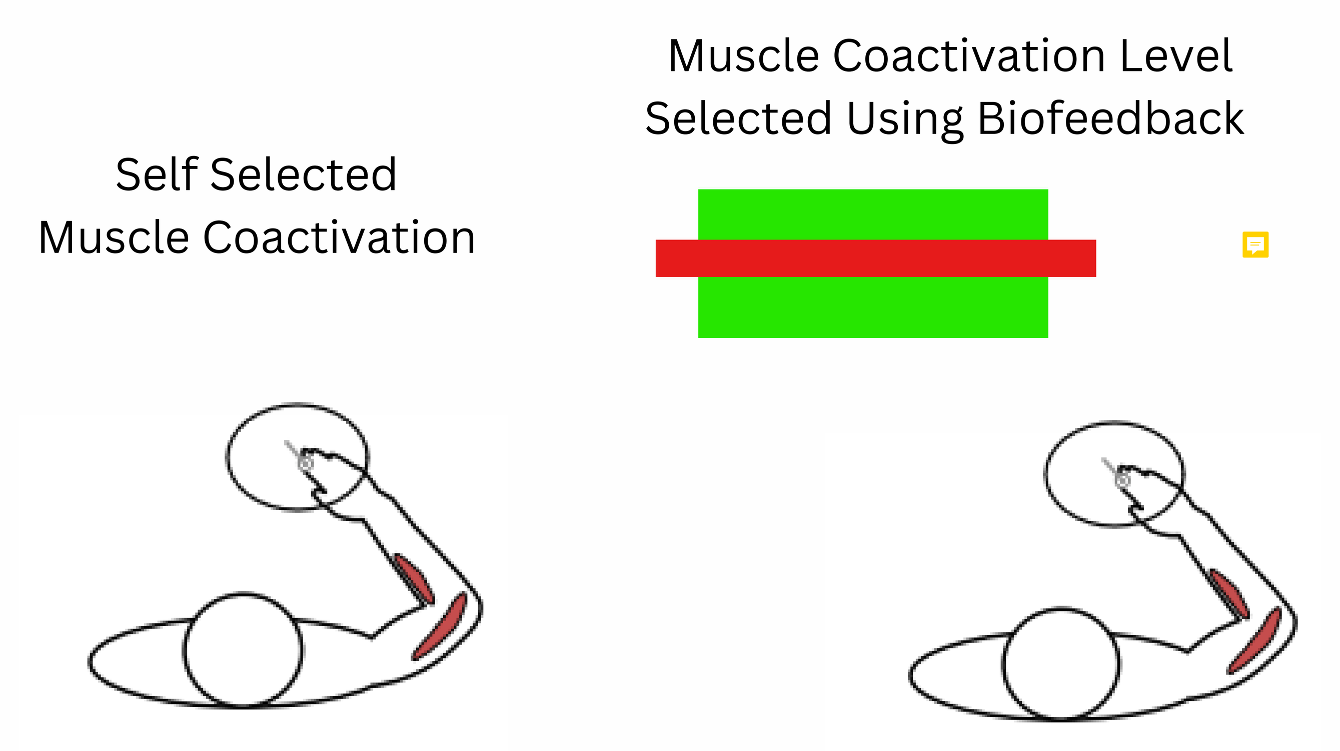
Muscle coactivation refers to the simultaneous activation of pairs of agonist and antagonist muscles. A longstanding and impactful idea in systems neuroscience is that coactivation stabilizes the body by making muscles stiff and resistant to motion to minimize errors caused by physical perturbations. Recent observations suggest that coactivation may improve performance by making the nervous system more responsive to sensory feedback. Muscle activity is costly, however, and we know little about the metabolic cost of coactivating muscles. Our research is crucial as there is a paucity of data on the trade-off between muscle coactivation, metabolic cost, and performance, highlighting the need for further investigation in this area. The primary objective of this study is to investigate the trade-off between metabolic cost and performance under different levels of muscle coactivation.

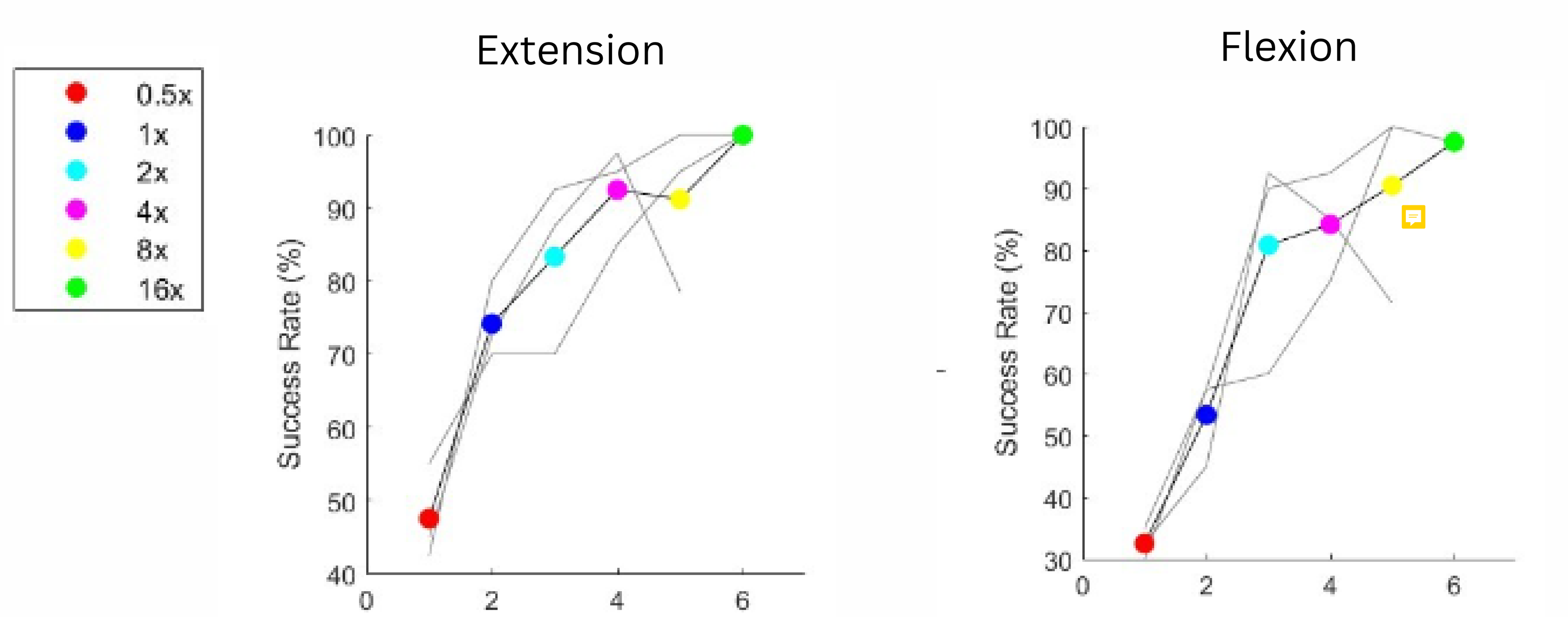
Hypothesis & Prediction

We hypothesize that healthy participants will naturally choose higher levels of coactivation for tasks that demand quicker responses to sensory feedback. We predict that participants will select higher levels of muscle coactivation when performing tasks that require rapid responses to mechanical disturbances.

Success rate (performance) associated with different levels of muscle coactivation

Five healthy participants (n=5) attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances. The disturbances rapidly extended or flexed the elbow and displaced the participant's arm from the target. They were required to return to the target within 400 ms. Participants first performed the task under self-selected levels of muscle coactivation. The activity of upper limb muscles using surface EMG, and behavioral responses using success rates. Biofeedback was then used to maintain a level of coactivation corresponding to [0.5, 2, 4, 8, 16]x their self-selected (1x) level.

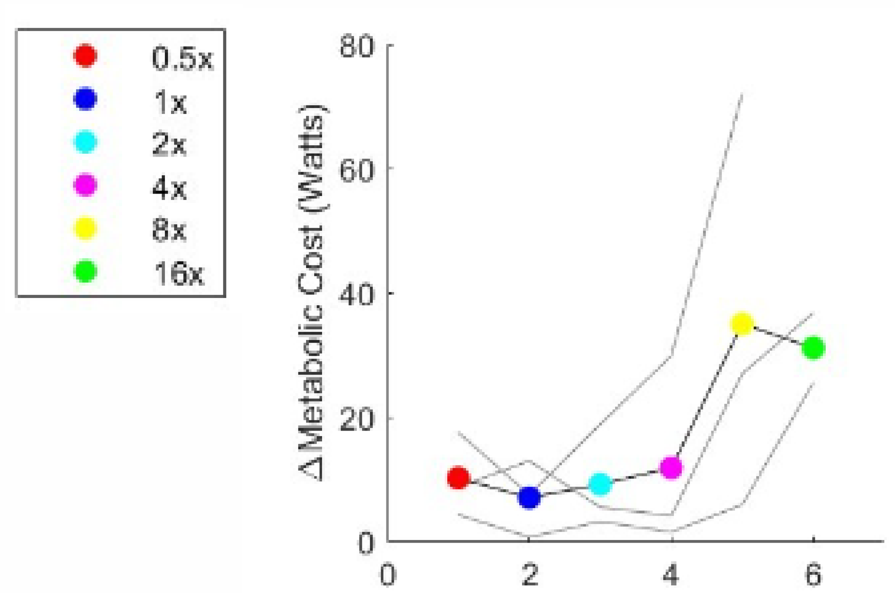




At self-selected levels of coactivation, participants had a success rate of 30-50%. We found that muscles became more responsive to the disturbances at higher coactivation levels, which limited the displacement of the arm, improved success rates to nearly 100%.

Metabolic costs associated with different levels of muscle coactivation

As muscle coactivation levels increased, the metabolic cost also rose significantly. This was measured using gas respirometry to monitor oxygen consumption and carbon dioxide production.



At the lowest coactivation level (0.5x), metabolic cost was minimal, while at the highest coactivation level (16x), metabolic cost peaked. This indicates that higher muscle coactivation, while beneficial for performance, demands greater energy expenditure, highlighting a clear trade-off between metabolic cost and enhanced responsiveness to mechanical disturbances.

Discussion

Higher levels of muscle coactivation significantly enhance performance by improving the responsiveness to mechanical disturbances. However, this increased performance comes at the cost of higher metabolic expenditure, suggesting a trade-off that participants manage by selecting coactivation levels that balance metabolic cost and task success.

Conclusion

Participants naturally prioritize metabolic cost over performance when selecting muscle coactivation levels, highlighting the complex interplay between metabolic cost and performance in human movement optimization.

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